

SPACEGAME

Design

1. Inspirations

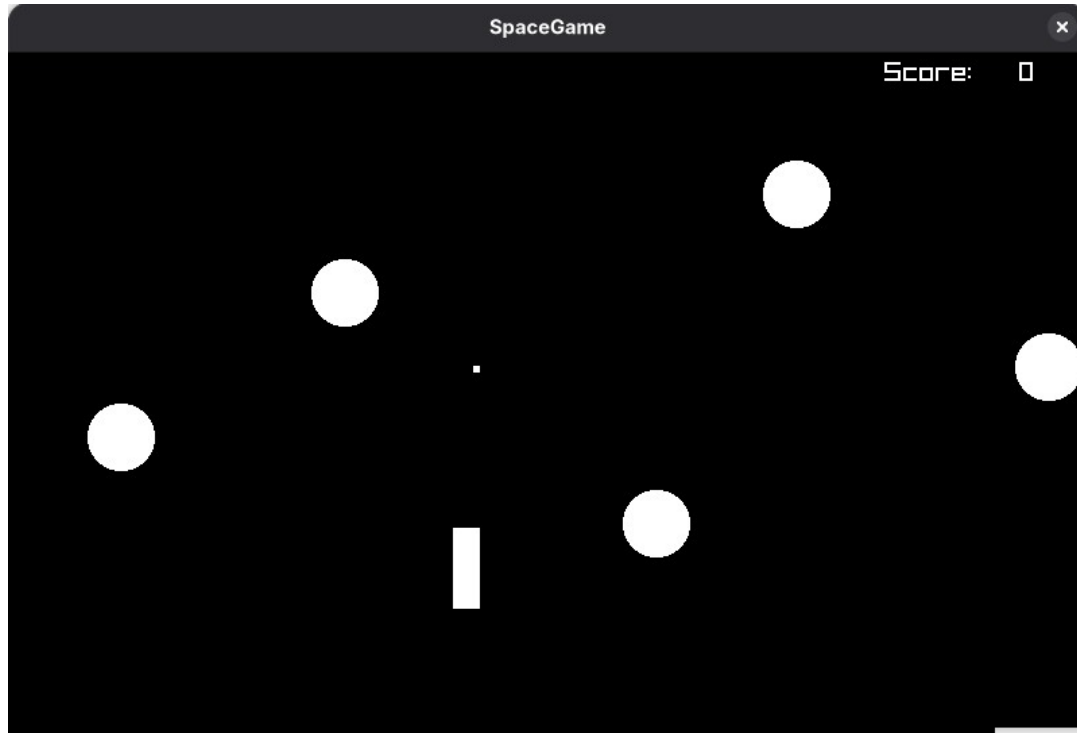


[Asteroids \(video game\) – Wikipedia](#)

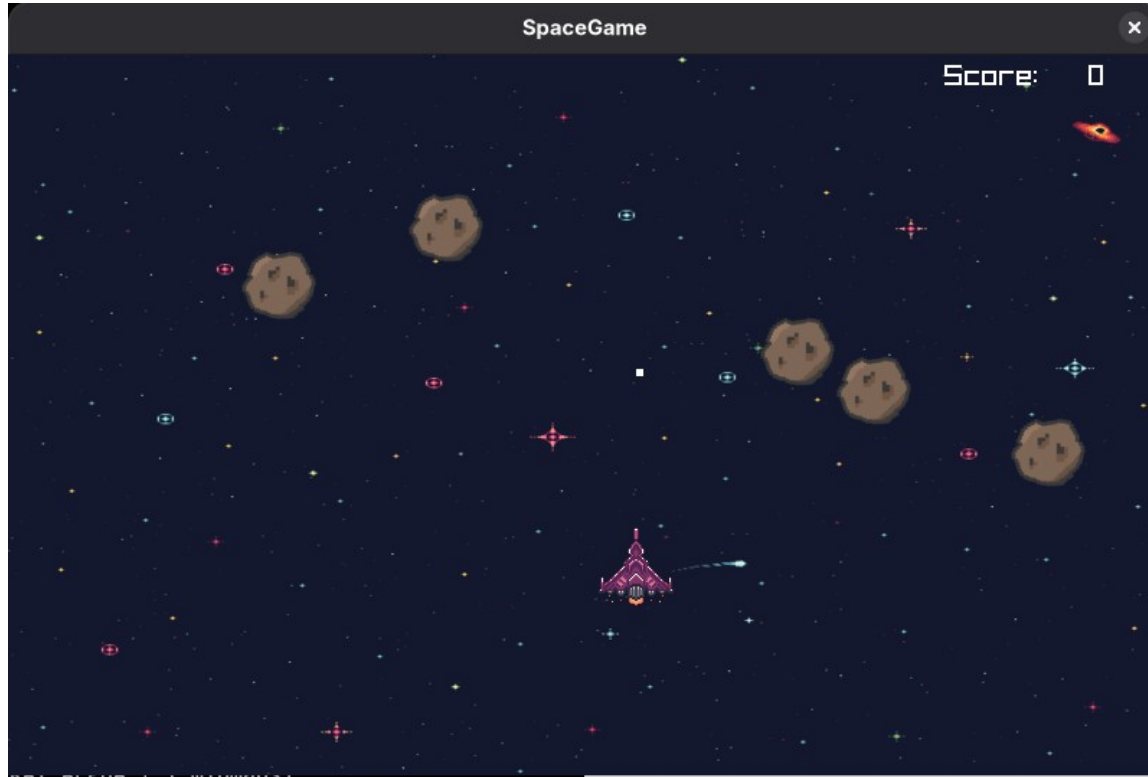


[Third generation of video game consoles – Wikipedia](#)

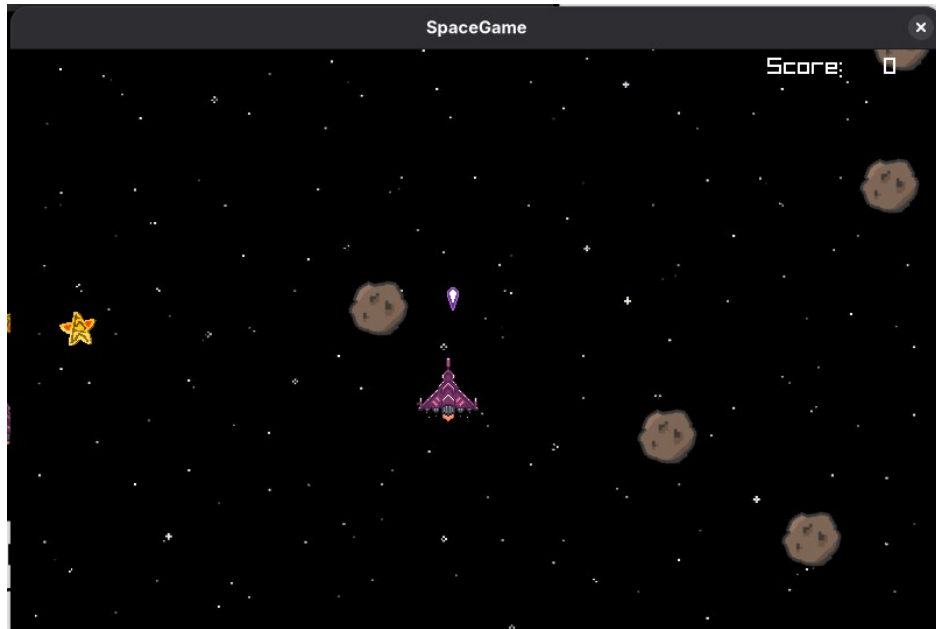
2. Prototypes



1. Early prototype - The image represents the ship as a rectangle, the asteroids as circles, the projectile as a square and the placement of the score counter.



2. Prototype with graphics – static background and images



3. Image showing the reward design and a different background

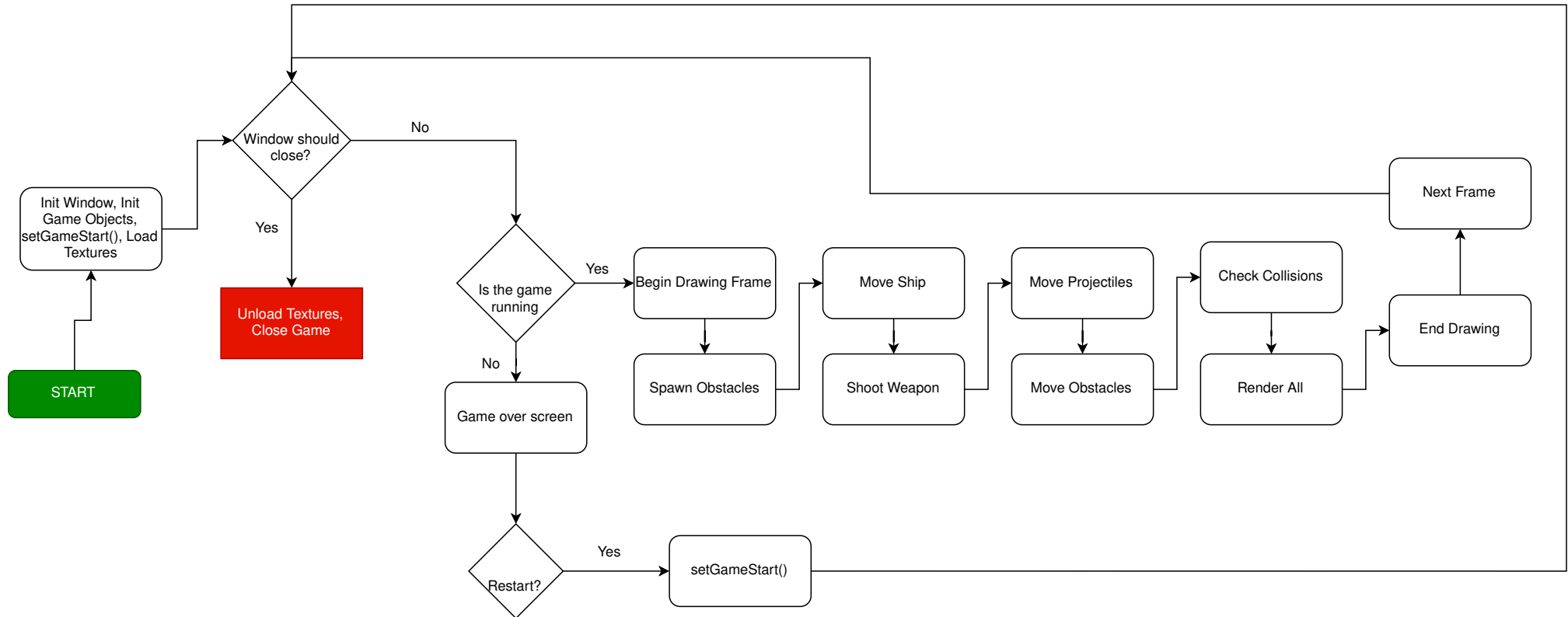


4. Image showing the Game Over menu



3. System Design

1. Flowchart of the game



4. Game Objects

Object	Data stored	Behaviour
Player (Player)	Position, dimensions, speed, alive state, angle	Moves with WASD/arrows; faces movement direction
Asteroid (Obstacle)	Position, radius, dimensions, speed, alive state	Moves downwards; respawns when inactive
Projectile (Projectile)	Position, dimensions, alive state	Fires with Space; travels upwards
Reward (Obstacle)	Position, radius, dimensions, speed, alive state	Moves downwards; grants points when collected